

Part A · Multiple Choice

2 points each

1. d) Acquire geographic resources

p. 9 — the four steps are ask, acquire, explore, analyze.

2. d) the speed of light

p. 19 — 299,792,458 meters per second, timed by the satellite's atomic clock. Because light is so fast, tiny timing errors become large position errors, which is why four satellites are used instead of three.

3. c) 380,000 years

p. 26 — protons, neutrons, and electrons formed within seconds; hydrogen within minutes; but neutral atoms took far longer.

4. d) balance the inward force of gravity against the outward force of nuclear fusion

p. 35 — the more massive the star, the more hydrogen it must burn to resist collapse, which is why massive stars run out of fuel sooner.

5. a) January 1st

p. 42 — at that scale each day is 12.5 million years and each second is 145 years. The dinosaurs go extinct on December 26th.

6. c) the extinction of the non-avian dinosaurs

p. 46 — the text notes this is no coincidence: most period and epoch boundaries are fixed by significant changes in the fossil record.

7. a) Extent is the area visible on the map; resolution is the smallest unit mapped

p. 55 — balancing the two is one of the most difficult decisions a cartographer makes.

8. c) compromise projection

p. 63 — rather than preserving any single property, it minimizes distortion of all of them, which suits general-purpose world maps.

9. a) About 7%

p. 68 — about 44% is visible light and the remainder is infrared, the longest wavelength, which we feel as heat.

10. c) September 22nd

p. 71 — the vernal equinox falls around March 21st or 22nd. On both, the Sun's rays shine most directly at the equator and day and night are equal.

11. c) The mesosphere

p. 82 — which is why air pressure there is so low that an unprotected traveler's blood would boil at normal body temperature.

12. d) a high-pressure zone

p. 89 — the high pressure forces winds downhill and clockwise (the Coriolis effect), so pressure and temperature rise while humidity falls. They often arrive at the end of California's dry season, raising fire danger.

13. a) the horizontal flow of air

p. 85 — warm air rises and creates low pressure, then flows horizontally at the top of the troposphere before cooling and descending.

- 14. c) mid-latitude cyclones that strike the mid-Atlantic and New England states**
p. 108 — named for the direction they come from. About 30 strike the region each year.
- 15. d) 157 mph and above**
p. 109 — both Category 4 and Category 5 are described as producing catastrophic damage.
- 16. d) an atom that has gained one or more electrons**
p. 117 — gaining negatively charged electrons gives a net negative charge. An atom that loses electrons becomes a positively charged cation.
- 17. a) cleavage**
p. 122 — the smooth surfaces and the predictable shape are the giveaway. Fracture produces breaks that are not controlled by the mineral's structure.
- 18. b) xenoliths**
p. 128 — Greek for “strange rocks.” The process of breaking them off is called stoping.
- 19. a) lava cools so rapidly that crystals do not develop at all**
p. 130 — the result is a glass. Pumice, by contrast, is the rock full of trapped gas-bubble holes, light enough to float.
- 20. b) granite and rhyolite**
p. 132 — gabbro and basalt are the mafic pair. In both cases the first rock named is the slow-cooling intrusive form and the second is the fast-cooling extrusive form.
- 21. b) shale**
p. 139 — without that bedding it is mudstone; if clay dominates the composition it is claystone. All are at least 75% silt and clay.
- 22. b) magnesium replacing some of the calcium in calcite**
p. 142 — this process, dolomitization, is thought to occur where magnesium-rich water percolates through carbonate tidal-flat sediments. Marine organisms do not make dolomite.
- 23. b) tilted in the same direction, with their upper ends pointing downstream**
p. 146 — that orientation is the most stable position with respect to the streamflow, so imbrication records flow direction.
- 24. a) migmatite**
p. 155 — it represents the upper limit of metamorphism, where the rock begins crossing back into the igneous part of the rock cycle.
- 25. b) retrograde metamorphism**
p. 161 — “retrograde” because the metamorphism happens at 200–300°C, well below the roughly 1200°C at which the rock originally formed. Greenstone is the non-foliated form; greenschist the foliated one.

Part B • Matching

8 points each

- 26. 1-b • 2-d • 3-g • 4-f • 5-h • 6-e • 7-c • 8-a**
1 point each. Analemma p. 72 • Coriolis effect p. 90 • Dew point p. 96 • Katabatic wind p. 88 • Occluded front p. 103 • Ionosphere p. 82 • Specific heat p. 85 • Tangent point p. 58.
- 27. 1-d • 2-f • 3-a • 4-h • 5-e • 6-g • 7-b • 8-c**
1 point each. Batholith p. 129 • Sill p. 129 • Arkose p. 140 • Breccia p. 140 • Evaporite p. 143 • Phyllite p. 154 • Hornfels p. 157 • Streak p. 119.

28. Because the data collected often raises new questions, which changes the order of the steps.

Rubric: 2 pts — the order of steps changes because more questions arise from the data that is collected; experiments that do or do not support a hypothesis lead to further questions and further experiments. 1 pt — nevertheless, logical and repeatable steps must be followed to reach verifiable conclusions. Credit the point that a hypothesis shown to be wrong is not a failed experiment, since all experimental results contribute to knowledge. (pp. 10 and 13)

29. Redshift is the shift of a star's dark absorption bands toward the red end of the spectrum, which occurs when the light source is moving away from the observer. Because every distant galaxy shows it, the universe must be expanding.

Rubric: 1 pt for what it is — absorption lines appear in the expected pattern but shifted toward the red end of the spectrum. 1 pt for the cause — the source is moving away, or the space between observer and source is being stretched. 1 pt for the conclusion — all galaxies show redshift, so all galaxies are moving away from each other and the universe is expanding. Credit the Doppler/ambulance-siren analogy. (p. 24)

30. In an undisturbed sequence of sedimentary layers, the layers at the bottom are the oldest and those above are younger. It fails where tectonic processes have overturned the sequence.

Rubric: 2 pts for the principle stated correctly, including the “otherwise undisturbed” condition. 1 pt for a valid exception — the text specifies a sequence that has been turned over by tectonic processes. Accept other well-reasoned answers involving tectonic disturbance, folding, or overthrusting. (pp. 47 and 144)

31. Mercator is a conformal projection: it preserves shape and angle, and the cost of doing so is severe distortion of area, which grows worse toward the poles.

Rubric: 1.5 pts — Mercator is conformal, so shapes and angles are accurate on all parts of the map. 1.5 pts — preserving shape strongly distorts area, and the distortion increases with distance from the tangent line, so landmasses near the poles are inflated relative to those near the equator. Credit reference to distortion ellipses: on a conformal map they vary greatly in size but all stay circular. (p. 60)

32. Magma cools and crystallizes into igneous rock; uplift and erosion expose it; weathering breaks it down; the fragments are eroded, transported, and deposited as sediment; burial compacts them and cementation lithifies them into sedimentary rock.

Rubric: 3 pts, roughly 0.5 per stage. (1) Magma cools — slowly at depth to form intrusive rock, or quickly at the surface to form extrusive rock. (2) Plate-tectonic mountain building uplifts the rock and erosion of overlying material exposes it. (3) Weathering breaks it down both physically and chemically. (4) The fragments are eroded and transported by glaciers, streams, waves, or wind. (5) They are deposited as sediments in rivers, lakes, deserts, or the ocean. (6) Burial compacts them and cementation lithifies them into solid sedimentary rock. (pp. 126–127, 135)

- 33. Foliation is a directional fabric in a metamorphic rock, produced when minerals grow or align perpendicular to the direction of squeezing. Its presence means the rock formed under directed pressure or shear stress.**

Rubric: 1.5 pts for what it is — a directional fabric; under directed pressure the rock deforms and minerals become elongated, or new minerals grow with their long axes perpendicular to the main stress, largely obliterating original bedding. 1.5 pts for what it indicates — the rock formed where pressure was *not* equal in all directions, typical of plate convergence and the roots of mountain ranges. Award credit for noting the exception: quartzite and marble may form under directed pressure yet show no foliation, because quartz and calcite do not align. (pp. 153–154)

Part D · Extended Response

8 points each

- 34. See rubric.**

Rubric: 2 pts — starting point. The grain sits in granite, an intrusive (plutonic) igneous rock formed when magma cooled slowly at depth, giving it a phaneritic texture with visible crystals. A batholith is an irregular pluton with more than 100 km² exposed.

2 pts — exposure and weathering. Plate-tectonic mountain building uplifts the batholith, and erosion of the overlying rock exposes it. At the surface it is weathered both physically (mechanical breaking) and chemically. Quartz survives this better than any other common mineral, which is why most sand is quartz.

2 pts — transport, deposition, lithification. The grain is eroded and transported by water, wind, ice, or gravity, moving by traction, saltation, or suspension depending on flow velocity. Where the current slows it is deposited, then buried and compacted, and cementation crystallizes minerals in the pores — lithification. The result is a sandstone (a quartz arenite if it is more than 90% quartz).

2 pts — metamorphism. Deeper burial subjects the sandstone to higher temperature and pressure. At medium to high grade it becomes quartzite, with the original grains welded together by additional silica. Full credit if the student notes quartzite is typically non-foliated because quartz does not align.

Award full credit for any coherent, correctly sequenced path; a student who routes the grain through a schist or gneiss instead should be credited if the parent rock and grade are consistent. (pp. 126–135, 156–157)

35. See rubric.

Rubric: Award up to 2.5 pts for each of three well-developed reasons, plus up to 0.5 pt for overall organization. Expect three of the following:

Gases for life. Plants use CO₂ and release O₂ through photosynthesis, which accounts for nearly all atmospheric oxygen; animals use that oxygen in respiration. Nitrogen and oxygen together are about 99% of the atmosphere.

Temperature moderation. Greenhouse gases — carbon dioxide, water vapor, methane, and ozone — trap heat radiating from the ground, insulating the planet. Without them temperatures would be frigid at night and scorching by day, as on the Moon. The gases differ in potency: one methane molecule traps 23 times the heat of one CO₂ molecule.

UV shielding. Ozone in the stratosphere, at roughly 15–30 km, absorbs most harmful ultraviolet radiation — all UVC, most UVB, and some UVA. High-energy UV penetrates cells and damages DNA, and without this shield most complex life could not survive.

Heat redistribution. Convection cells and the global wind belts move heat from the equator toward the poles, evening out temperatures that would otherwise be unlivably extreme at both ends. Cold air masses flow toward the equator and warm air masses toward the poles.

Water cycling. The atmosphere carries water vapor and delivers it as precipitation, driving the hydrological cycle that keeps the rock cycle running and sustains life on land.

Full credit requires a specific mechanism named for each reason, not just a general claim that the atmosphere is important. (pp. 77, 81, 90–93, 126)